

- Q.26 Describe the Watershed Management. (CO3)
- Q.27 What are factors affecting the erosion by water? (CO2)
- Q.28 Differentiate between Surface and Sub-Surface drainage. (CO6)
- Q.29 Describe the reclamation of Alkaline Soil. (CO5)
- Q.30 Explain the use of bunds to control erosion. (CO3)
- Q.31 Describe the gravimetric methods of soil moisture determination. (CO1)
- Q.32 Define Soil Texture. Write USDA classification of soil texture. (CO1)
- Q.33 Briefly describe the different systems of water harvesting. (CO4)
- Q.34 Describe in brief the different practices adopted to control erosion. (CO3)
- Q.35 Discuss in brief the mechanics of wind erosion. (CO2)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Write a detailed note on the problem of Water logging Describe its causes effects and management. (CO6)
- Q.37 Explain the principle of gully erosion and give the detailed classification of gullies. (CO2)
- Q.38 Describe the principles, adaptability, and constructional features of Drop Spillway. (CO3)

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M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Wind erosion is very significant in (CO2)
- a) Clay soil b) Loamy soil
- c) Sandy soil d) Black cotton soil
- Q.2 Darcy's Law is applicable when Renolds number is (CO3)
- a) Less than 1 b) Equal to 1
- c) More than 1 d) Less than 2
- Q.3 The method of rainwater harvesting which can be adopted by individual house owners is (CO4)
- a) Construction of recharge trenches
- c) On channel storage of water
- d) Creation of new water bodies
- d) Roof-top rainwater harvesting
- Q.4 Which soil has high water retaining capacity(CO1)
- a) Sand b) Silt
- c) Clay d) Loam

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- Q.5 Which of the following drop structure can also be used for water storage along with gully control. (CO3)
- Straight drop structure
 - Drop inlet spillway
 - Oogee spillway
 - Chute spillway
- Q.6 Acid soils can be reclaimed by (CO1)
- CaCO_3
 - $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
 - H_2SO_4
 - HNO_3
- Q.7 Which of the following is not related to farm pond (CO5)
- Fish farming
 - Rain water management
 - Rainfall farming
 - Soil erosion
- Q.8 The relation between void ratio and porosity. (CO1)
- $n = (1+e)/e$
 - $n = e/(1+e)$
 - $e = n/(1+n)$
 - None of the above
- Q.9 In roof water harvesting system the catchment area is the (CO4)
- Rainfall volume
 - Gutter surface
 - Roof area X runoff
 - roof area receiving the rainfall, directly
- Q.10 In Chute spillway, the flow is usually (CO3)
- Uniform
 - Sub critical
 - Critical
 - Super critical

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What do you mean by Strip Cropping. (CO3)
- Q.12 The maximum moisture is available to plant at _____. (CO1)
- Q.13 Write the formula of Particle Density. (CO1)
- Q.14 SAR stands for _____. (CO5)
- Q.15 Define reclamation of soil. (CO5)
- Q.16 Define drainage coefficient. (CO6)
- Q.17 What do you mean by Strip Cropping (CO3)
- Q.18 Define water logging. (CO6)
- Q.19 Define sheet erosion. (CO2)
- Q.20 Chute spillways are used at _____ slope. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are different soils separates? Give one classification of soil separates. (CO1)
- Q.22 What are the different reasons for the formation of saline, alkaline and acid soil? (CO5)
- Q.23 Discuss the soil conservation through tree and grass cultivation. (CO3)
- Q.24' What is ground water recharge? Explain its importance. (CO4)
- Q.25 Define soil permeability and its importance. (CO1)

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